DETAILS OF CLAUDE'S $h = 2$ ARGUMENT

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Say $h_0 = 1$ and $h_n = 2$ for n positive. From Lemma 2.1, with $H(t) = \sum_{n\geq 0} h_n t^n$ and $J(t) = \sum_{r\geq 1} j_r t^r$, the following statements are equivalent:

(i) $tH'(t) = H(t)\,J(t)$,
(ii) $nh_n = \sum_{r=1}^{n} j_r\,h_{n-r} \quad (n \geq 1)$.

From (a), we also have
(iii) $J(t) = t\,\frac{d}{dt}\log H(t)$.

With $h_0 = 1$ and $h_r = 2$ for $r \geq 1$,

$$H(t) = 1 + 2\sum_{r\geq 1} t^r = 1 + \frac{2t}{1-t} = \frac{1+t}{1-t}.$$

Hence

$$\frac{d}{dt}\log H(t) = \frac{d}{dt}\big(\log(1+t) - \log(1-t)\big) = \frac{1}{1+t} + \frac{1}{1-t} = \frac{2}{1-t^2},$$

so

$$J(t) = t\cdot\frac{2}{1-t^2} = 2t + 2t^3 + 2t^5 + \cdots .$$

so $j_r = 2$ for odd r and $j_r = 0$ for even r.

Writing $\{j_n^{(c)}\}_{n\geq 1} = \{c, j_1, j_2, \dots\}$ and $J^{(c)}(t) = \sum_{r\geq 1} j_r^{(c)} t^r$,

$$J^{(c)}(t) = c\,t + \sum_{r\geq 2} j_{r-1}t^r = c\,t + t\sum_{s\geq 1} j_s t^s = c\,t + t\,J(t) = c\,t + t\cdot\frac{2}{1-t^2}.$$

Let the deformed sequence $\{h_n^{(c)}\}_{n\geq 0}$ be paired with $\{j_n^{(c)}\}_{n\geq 1}$ through the relation (D) and substitute $H^{(c)}(t)$ for $H(t)$ in (i) - (iii). By (iii),

$$\frac{d}{dt}\log H^{(c)}(t) = \frac{J^{(c)}(t)}{t} = c + \frac{2t}{1-t^2}.$$

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Next we evaluate the anti-derivative $I = \displaystyle\int \frac{2t}{1-t^2}\,dt$ in a neighborhood of zero so small that $1-t^2 > 0$. Making the substitution $u = 1-t^2$, $du = -2t\,dt$,

$$I = \int \frac{2t}{1-t^2}\,dt = -\int \frac{du}{u} = -\log u + C = -\log(1-t^2) + C,$$

so

$$\log H^{(c)}(t) = c\,t - \log(1-t^2) + C.$$

From $h_0^{(c)} = 1$ and $H^{(c)}(0) = 1$, we have $\log H^{(c)}(0) = 0$, and $C = 0$. Therefore

$$H^{(c)}(t) = \frac{e^{ct}}{1-t^2}.$$

To expand $H^{(c)}(t)$ in powers of t, let us write

$$\sum_{n=0}^{\infty} h_n^{(c)} t^n = H^{(c)}(t) = e^{ct}/(1-t^2) = \sum_{a=0}^{\infty} \frac{c^a}{a!} t^a \times \sum_{b=0}^{\infty} t^{2b} =$$

$$\sum_{a=0}^{\infty} \frac{c^a}{a!} t^a \times \sum_{b=0}^{\infty} \gamma_b t^b,$$

where γ_b is 0 for b odd and is 1 for b even. So

$$h_n^{(c)} = \sum_{\substack{a+b=n\\ a\geq 0\\ b\geq 0}} \frac{c^a}{a!}\gamma_b = \sum_{a=0}^{n} \frac{c^a}{a!}\gamma_{n-a}.$$

For even n, this becomes

$$h_n^{(c)} = \sum_{\substack{0\leq a\leq n\\ a\equiv n \mod 2}} \frac{c^a}{a!},$$

and this approaches $\cosh c$ as $n \to \infty$ over even n. Similarly, $h_n^{(c)}$ approaches $\sinh c$ as $n \to \infty$ over odd n.

We give a self-contained analytic derivation of the large-n behavior of the coefficients $h_n^{(c)}$ defined by the deformed generating function $H^{(c)}(t) = e^{ct}/(1-t^2)$. The only input is the closed form on the unit disc; no individual coefficient is computed and no numerical evidence is invoked. We obtain an exact formula for $h_n^{(c)}$, an explicit bound on the error in the limiting expression, and the parity-split limits $\cosh c$ and $\sinh c$.

1. Setup and notation

Fix $c \in \mathbb{C}$. We take as our single starting point the identity, valid as an equality of analytic functions on the open unit disc $\{\, t \in \mathbb{C} : |t| < 1 \,\}$,

$$H^{(c)}(t) \;=\; \frac{e^{ct}}{1-t^2} \;=\; \sum_{n=0}^{\infty} h_n^{(c)}\, t^n. \tag{1}$$

This is what equation (D) produces for the constant sequence after deformation: solving $\frac{d}{dt}\log H^{(c)}(t) = c + \frac{2t}{1-t^2}$ with $H^{(c)}(0) = 1$ yields $\log H^{(c)}(t) = ct - \log(1-t^2)$ on a neighborhood of $t = 0$, hence (1), and the two sides, being analytic on the unit disc, agree throughout it. The numbers $h_n^{(c)}$ are exactly the Taylor coefficients in (1); their large-n behavior is the object of study. Nothing below uses the value of any particular $h_n^{(c)}$.

For $z \in \mathbb{C}$ and $n \geq 0$ write

$$E_n(z) \;=\; \sum_{k=0}^{n} \frac{z^k}{k!}, \qquad R_n(z) \;=\; e^z - E_n(z) \;=\; \sum_{k=n+1}^{\infty} \frac{z^k}{k!},$$

so that $E_n(z)$ is the n-th Taylor polynomial of exp at 0 and $R_n(z)$ is its remainder.

2. An exact formula for the coefficients

Lemma 2.1. *For every $c \in \mathbb{C}$ and every $n \geq 0$,*

$$h_n^{(c)} \;=\; \tfrac{1}{2}\, E_n(c) \;+\; \tfrac{(-1)^n}{2}\, E_n(-c).$$

Proof. The partial-fraction identity

$$\frac{1}{1-t^2} \;=\; \frac{1}{2}\left(\frac{1}{1-t} + \frac{1}{1+t}\right),$$

an equality of rational functions on $\mathbb{C} \setminus \{\pm 1\}$, gives on the disc $|t| < 1$

$$H^{(c)}(t) \;=\; \frac{1}{2}\,\frac{e^{ct}}{1-t} \;+\; \frac{1}{2}\,\frac{e^{ct}}{1+t}.$$

For $|t| < 1$ the exponential series $e^{ct} = \sum_{k\geq 0} \frac{c^k}{k!} t^k$ converges absolutely (it is entire), and the geometric series $\frac{1}{1\mp t} = \sum_{i\geq 0} (\pm 1)^i t^i$ converges absolutely. The product of two absolutely convergent power series is represented on the common disc of convergence by their Cauchy product, which again converges absolutely; hence the coefficient of t^n may be read off as a finite convolution. Thus

$$[t^n]\,\frac{e^{ct}}{1-t} = \sum_{k=0}^{n} \frac{c^k}{k!} = E_n(c),$$

$$[t^n]\,\frac{e^{ct}}{1+t} = \sum_{k=0}^{n} \frac{c^k}{k!}\,(-1)^{n-k} = (-1)^n \sum_{k=0}^{n} \frac{(-c)^k}{k!} = (-1)^n E_n(-c).$$

Since the Taylor coefficients of the analytic function $H^{(c)}$ are unique, comparison with (1) yields the stated identity. $\square$

3. A tail estimate for the exponential

Lemma 3.1. *For every $z \in \mathbb{C}$ and every $n \geq 0$,*

$$\big|R_n(z)\big| \;=\; \big|e^z - E_n(z)\big| \;\leq\; \frac{|z|^{\,n+1}}{(n+1)!}\, e^{|z|}.$$

Proof. Estimating the tail term by term,

$$\big|R_n(z)\big| \;\leq\; \sum_{k=n+1}^{\infty} \frac{|z|^k}{k!} \;=\; \frac{|z|^{\,n+1}}{(n+1)!} \sum_{j=0}^{\infty} \frac{(n+1)!}{(n+1+j)!}\, |z|^{\,j}.$$

For each $j \geq 0$,

$$\frac{(n+1)!\, j!}{(n+1+j)!} \;=\; \binom{n+1+j}{j}^{-1} \;\leq\; 1, \qquad \text{so} \qquad \frac{(n+1)!}{(n+1+j)!} \;\leq\; \frac{1}{j!}.$$

Therefore

$$\big|R_n(z)\big| \;\leq\; \frac{|z|^{\,n+1}}{(n+1)!} \sum_{j=0}^{\infty} \frac{|z|^{\,j}}{j!} \;=\; \frac{|z|^{\,n+1}}{(n+1)!}\, e^{|z|}. \qquad \square$$

4. The asymptotics

Theorem 4.1. *For every $c \in \mathbb{C}$ and every $n \geq 0$,*

$$h_n^{(c)} \;=\; \frac{e^c + (-1)^n e^{-c}}{2} \;+\; \varepsilon_n(c), \qquad \big|\varepsilon_n(c)\big| \;\leq\; \frac{|c|^{\,n+1}}{(n+1)!}\, e^{|c|}.$$

Consequently $\varepsilon_n(c) \to 0$ as $n \to \infty$, faster than any geometric sequence, and

$$\lim_{\substack{n\to\infty \\ n \text{ even}}} h_n^{(c)} = \frac{e^c + e^{-c}}{2} = \cosh c, \qquad \lim_{\substack{n\to\infty \\ n \text{ odd}}} h_n^{(c)} = \frac{e^c - e^{-c}}{2} = \sinh c.$$

Proof. Substitute $E_n(\pm c) = e^{\pm c} - R_n(\pm c)$ into Lemma 2.1:

$$\begin{aligned} h_n^{(c)} &= \tfrac{1}{2}\big(e^c - R_n(c)\big) + \tfrac{(-1)^n}{2}\big(e^{-c} - R_n(-c)\big) \\ &= \frac{e^c + (-1)^n e^{-c}}{2} \;-\; \tfrac{1}{2}\big(R_n(c) + (-1)^n R_n(-c)\big). \end{aligned}$$

Define $\varepsilon_n(c) = -\frac{1}{2}\big(R_n(c) + (-1)^n R_n(-c)\big)$. Because $|-c| = |c|$, Lemma 3.1 bounds each of $|R_n(c)|$ and $|R_n(-c)|$ by $\dfrac{|c|^{\,n+1}}{(n+1)!}\, e^{|c|}$, so

$$\big|\varepsilon_n(c)\big| \leq \tfrac{1}{2}\big(|R_n(c)| + |R_n(-c)|\big) \leq \frac{|c|^{\,n+1}}{(n+1)!}\, e^{|c|}.$$

The quantity $|c|^{n+1}/(n+1)!$ is the general term of the convergent series for $e^{|c|}$, so it tends to 0; moreover

$$\frac{|c|^{\,n+2}/(n+2)!}{|c|^{\,n+1}/(n+1)!} = \frac{|c|}{n+2} \xrightarrow[n\to\infty]{} 0,$$

which is the meaning of "faster than any geometric sequence." Fixing the parity of n makes $(-1)^n$ constant, and letting $n \to \infty$ through that parity class sends $\varepsilon_n(c) \to 0$, giving the two limits. $\square$

Corollary 4.2 (the case $c = 1$)**.** *For $c = 1$,*

$$h_n^{(1)} = \frac{e + (-1)^n e^{-1}}{2} + \varepsilon_n, \qquad |\varepsilon_n| \le \frac{e}{(n+1)!},$$

so $h_n^{(1)} \to \cosh 1 = \dfrac{e}{2} + \dfrac{1}{2e}$ over even n and $h_n^{(1)} \to \sinh 1 = \dfrac{e}{2} - \dfrac{1}{2e}$ over odd n.

Proof. Set $c = 1$ in Theorem 4.1; then $|c| = 1$ gives the bound $\dfrac{1^{\,n+1}}{(n+1)!}\, e^1 = \dfrac{e}{(n+1)!}$, and $\frac{1}{2}(e \pm e^{-1}) = \cosh 1$ or $\sinh 1$ according to parity. $\square$

5. Remarks

Remark 1 (which pole supplies what)*.* The decomposition in Lemma 2.1 separates the two simple poles of $H^{(c)}$ at $t = \pm 1$. The pole at $t = +1$ contributes the parity-independent term $e^c/2$ (the limit of $E_n(c)$); the pole at $t = -1$ contributes the alternating term $(-1)^n e^{-c}/2$. The alternation $(-1)^n$, and hence the splitting of the limit into the two distinct values $\cosh c$ and $\sinh c$, is produced solely by the *location* $t = -1$ of the second pole and is independent of c. The parameter c enters only through the numerator e^{ct} and therefore fixes nothing but the two residue magnitudes $e^{\pm c}/2$.

Remark 2 (independence from numerics)*.* Every assertion follows from (1) together with the elementary Lemmas 2.1 and 3.1. In particular the limit is not an extrapolation from computed coefficients: for each finite n the deviation of $h_n^{(c)}$ from $\frac{1}{2}\big(e^c + (-1)^n e^{-c}\big)$ is bounded explicitly by $\dfrac{|c|^{\,n+1}}{(n+1)!}\, e^{|c|}$.

Remark 3 (consistency with the small-t derivation)*.* The identity (1) is established on a neighborhood of $t = 0$, and the asymptotics are extracted without ever evaluating $H^{(c)}$ at or near the poles. The geometric expansions $\frac{1}{1 \mp t} = \sum_{i \ge 0} (\pm 1)^i t^i$ used in Lemma 2.1 converge on exactly the disc $|t| < 1$ on which (1) holds. Thus the whole argument takes place inside the region where the derivation of (1) is valid; the role of the poles at $t = \pm 1$ is only to mark the boundary $|t| = 1$ of that disc, which is where the coefficient asymptotics are governed.

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